

GoodWe Modbus Protocol of Inverter

Grid-tied XS G3/DNS G3/MS G3/SDT G3 series (Customer Version)



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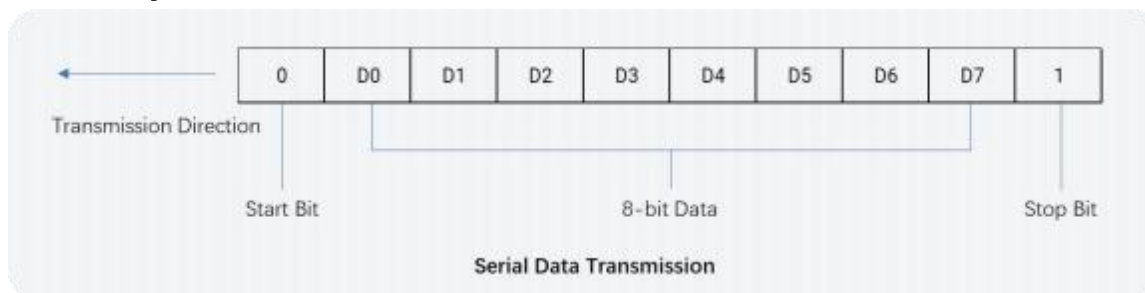
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Version change record table				
version number	date	editor	change log	Program Version
V 1.0	2023.02.08	frank	1. Initial version	
V 1.1	2023.03.01	frank	1. Modify the remark information of register 4 0481	
V 1.2	2023.05.25	frank	1.Add applicable model S DTG3 series	
V 1.3	2024 .02.05	Song.liu	1. Adjust the high/low order of CRC check bits; 2. Update debugging examples; 3. Adjust the modbus address setting instructions in Chapter 5; 4. Update the notes in Table 8-2; 5. Add 30035 , 40352, 40362	XSG3/DNSG3/MS G3 DSP : V1.00.00 ARM: V1.00
				SDT G3 DSP : V1.00.00 ARM: V1.00

This MODBUS RTU protocol applies to the communication between GoodWe Grid-tied inverter XS G3/DNS G3/MS G3/SDT G3 series and the PC monitoring software.

This protocol adopts MODBUS protocol RTU mode, which can be used to read real-time operating data and fault status of the inverter, and to perform remote control.

1. Byte Format



A byte is of 8 binary digits working together with 1 start bit (0) and 1 stop bit (1) in transmission, 10 bits in total. As illustrated below, D0 is the least significant bit (LSB) and D7 is the most significant bit (MSB). Pass to the least significant bit first, then to the most significant bit.

2. Communication Data Format

Data are exchanged in the form of word or doubleword:

Data Type	Register Size	Byte Size	Description
Byte Data	1	1	
Integer Data	1	2	One forwarding at one time, from the most significant bit to the least significant bit.
Long Integer Data	2	4	Two separate forwarding, from the most significant bit to the least significant bit.
Floating-point Data			

3. Frame Format

3.1 Read the Content of Register(Function code: 03H)

3.1.1 Frame format from host computer

No.	Code	Example	Description
1	Device Address	1	The device's communication address range 1-247
2	03H	03H	Function code
3	MSB of First Register	00H	The address of first register
4	LSB of First Register	01H	
5	MSB of Register Number	00H	Register number
6	LSB of register Number	02H	
7	CRC16 MSB of Checksum	95H	CRC Checksum Data
8	CRC16 LSB of Checksum	CBH	

3.1.2 Frame format by device (normal data)

No.	Code	Description
1	Device Address	The device's communication address range 1-247
2	03H	Function code
3	Byte Number of Receiving Data (2N)	Number of data
4	MSB of First Register Data	High bit of Data 1
5	LSB of First Register Data	Low bit of Data 1
...	...	...
2N+2	MSB of Register Data N	High bit of Data N
2N+3	LSB of Register Data N	Low bit of Data N
2N+4	CRC16 MSB of Checksum	CRC Checksum Data
2N+5	CRC16 LSB of Checksum	

3.1.3 Frame format by device (error of first register address or number)

No.	Code	Description
1	Device Address	The device's communication address range 1-247
2	83H	Function code
3	02H	Error Code
4	CRC16 MSB of Checksum	High bit of checksum
5	CRC16 LSB of Checksum	Low bit of checksum

3.2 Set the Content of Register (Function code: 10H)

3.2.1 Frame format from host computer

No.	Code	Example	Description
1	Device address	1	The device's communication address range 1-247
2	10H	10H	Function code
3	MSB of First Register	00H	Address of register 0000H
4	LSB of First Register	00H	
5	MSB of Register Number	00H	Register number 01H
6	LSB of Register Number	01H	
7	Byte Number (N)	02H	Byte number of register 02H
8	MSB of Data	0AH	Data 0AF0H
9	LSB of Data	F0H	
10	CRC16 MSB of Checksum	A0H	CRC Checksum
11	CRC16 LSB of Checksum	B4H	

3.2.2 Frame format by device (write successfully)

No.	Code	Example	Description
1	Device Address	1	The device's communication address range 1-247
2	10H	10H	Function code
3	MSB of First Register	00H	Address of register 0000H
4	LSB of First Register	00H	
5	MSB of Register Number	00H	Register number 01H
6	LSB of Register Number	01H	
7	CRC16 MSB of Checksum	01H	CRC Checksum
8	CRC16 LSB of Checksum	C9H	

3.2.3 Frame format by device (data over limit error)

No.	Code	Description
1	Device Address	The device's communication address range 1-247
2	90H	Function code
3	03H	Error code
4	CRC16 MSB of Checksum	CRC Checksum
5	CRC16 LSB of Checksum	

3.2.4 Frame format by device (error of register address or number)

No.	Code	Description
1	Device Address	The device's communication address range 1-247
2	90H	Function code
3	02H	Error code
4	CRC16 MSB of Checksum	CRC Checksum
5	CRC16 LSB of Checksum	

4. Baud Rate of 485Communication

9600bps

5. Device Address

The range of device address is 1 – 247 with 247 as default setting before delivery.

Please set the device address by yourself through SolarGo or the device display screen on site.

6. Communication Code

03H: read (Note: read of one or more consecutive addresses are supported.)

10H: write (Note: RTC time should be written together.)

7. CRC Checking of Communication

The CRC checking is performed from the first byte to the byte before the MSB of CRC checksum.

8. Register Address of Device

Table 8-1 register address and attributes of device

#Address Register address (decimal)	#Address Register address (hex)	English Name	#R/ W	#Type	#Size	#SF	#Units	Range	Note
30004	0x7534	Device Serial Number	RO	STR	8	1	N/A	[0, 0]	ASCII code, 16 bytes
30034	0x7552	DSP1 software version	RO	U16	1	1	N/A	[0, 0]	DSP1 software version
30035	0x7553	DSP2 software version	RO	U16	1	1	N/A	[0, 0]	DSP2 software version
30103	0x7597	PV1 voltage	RO	U16	1	10	V	[0, 0]	
30104	0x7598	PV1 current	RO	U16	1	10	A	[0, 0]	
30105	0x7599	PV2 voltage	RO	U16	1	10	V	[0, 0]	
30106	0x759A	PV2 current	RO	U16	1	10	A	[0, 0]	
30107	0x759B	PV3 voltage	RO	U16	1	10	V	[0, 0]	
30108	0x759C	PV3 current	RO	U16	1	10	A	[0, 0]	
30115	0x75A3	Phase voltage AB	RO	U16	1	10	V	[0, 0]	Don' t support single-phase models
30116	0x75A4	Phase voltage BC	RO	U16	1	10	V	[0, 0]	Don' t support single-phase models
30117	0x75A5	Phase voltage CA	RO	U16	1	10	V	[0, 0]	Don' t support single-phase models
30118	0x75A6	Phase voltage AN	RO	U16	1	10	V	[0, 0]	
30119	0x75A7	Phase voltage BN	RO	U16	1	10	V	[0, 0]	Don' t support single-phase models
30120	0x75A8	Phase voltage CN	RO	U16	1	10	V	[0, 0]	Don' t support single-phase models
30121	0x75A9	Phase A current	RO	U16	1	10	A	[0, 0]	
30122	0x75AA	Phase B current	RO	U16	1	10	A	[0, 0]	Don' t support single-phase models

30123	0x75AB	Phase C current	RO	U16	1	10	A	[0, 0]	Don' t support single-phase models
30124	0x75AC	Phase A frequency	RO	U16	1	100	Hz	[0, 0]	
30125	0x75AD	Phase A frequency	RO	U16	1	100	Hz	[0, 0]	Don' t support single-phase models
30126	0x75AE	Phase A frequency	RO	U16	1	100	Hz	[0, 0]	Don' t support single-phase models
30127	0x75AF	AC active power	RO	U32	2	1000	kW	[0, 0]	AC active power
30129	0x75B1	Inverter present status	RO	U16	1	1	N/A	[0, 0]	0:waiting 1:on grid 2:fault 3:off grid 4:self-check
30130	0x75B2	Fault code	RO	U32	2	1	N/A	[0, 0]	Table 8-2 for details
30132	0x75B4	Warning code	RO	U16	1	1	N/A	[0, 0]	
30133	0x75B5	AC apparent power	RO	U32	2	1000	kVA	[0, 0]	
30135	0x75B7	AC reactive power	RO	S32	2	1000	kVar	[0, 0]	
30141	0x75BD	Cabinet Temperature	RO	S16	1	10	°C	[0, 0]	
30144	0x75C0	Daily AC energy	RO	U16	1	10	kWh	[0, 0]	
30145	0x75C1	Gross AC energy	RO	U32	2	10	kWh	[0, 0]	
30147	0x75C3	Total generation time	RO	U32	2	1	H	[0, 0]	
30149	0x75C5	Safety Country Code	RO	U16	1	1	N/A	[0, 0]	
30163	0x75D3	BUS Voltage	RO	U16	1	10	N/A	[0, 0]	
30164	0x75D4	NBUS Voltage	RO	U16	1	10	N/A	[0, 0]	Don' t support single-phase models

30165	0x75D5	Derating Mode	RO	U32	2	1	N/A	[0, 0]	Bit 0: Overtemperature derating Bit 1: Active power derating Bit 2: Reactive power derating(PF/QU/FixQ) Bit 3: Power VS Frequency Bit 4: PU Curve Bit 5: Export power derating Bit 6: DRED derating Bit 7: Limited power start derating Bit 8: Maximum current derating Bit 9: Overvoltage derating (For GW) Bit 10: Power calibration limit power (For ATS)
33050	0x811A	SpecOverExtPowerFactor	RO	S16	1	100	N/A	[0, 0]	For North American 1547-1 requirements
33051	0x811B	SpecOverExtPower	RO	U16	1	1	W	[0, 0]	For North American 1547-1 requirements
33052	0x811C	SpecUnderExtPowerFactor	RO	S16	1	100	N/A	[0, 0]	For North American 1547-1 requirements
33053	0x811D	SpecUnderExtPower	RO	U16	1	1	W	[0, 0]	For North American 1547-1 requirements
33054	0x811E	ApparentPowerRate	RO	U16	1	1	W	[0, 0]	For North American 1547-1 requirements
33055	0x811F	NormOperCategory	RO	STR	8	1	N/A	[0, 0]	For North American 1547-1 requirements
33063	0x8127	AbnormOperCategory	RO	STR	8	1	N/A	[0, 0]	For North American 1547-1 requirements
33071	0x812F	ReactiveInjectPowerRate	RO	S16	1	1	VA	[0, 0]	For North American 1547-1 requirements

33072	0x8130	ReactiveAbsorbPowerRate	RO	S16	1	1	VA	[0, 0]	For North American 1547-1 requirements
33073	0x8131	ActivePowerRate	RO	U16	1	1	W	[0, 0]	For North American 1547-1 requirements
33074	0x8132	ApparentPowerRate	RO	U16	1	1	VA	[0, 0]	For North American 1547-1 requirements
33075	0x8133	ACVoltNormRate	RO	U16	1	10	V	[0, 0]	For North American 1547-1 requirements
33076	0x8134	ACVoltMaxRate	RO	U16	1	10	V	[0, 0]	For North American 1547-1 requirements
33077	0x8135	ACVoltMinRate	RO	U16	1	10	V	[0, 0]	For North American 1547-1 requirements
33078	0x8136	SupportedControlMode	RO	U32	2	1	N/A	[0, 0]	For North American 1547-1 requirements
33080	0x8138	Manufacturers	RO	STR	8	1	N/A	[0, 0]	For North American 1547-1 requirements
33088	0x8140	ReactiveSuptance	RO	U16	1	1	N/A	[0, 0]	For North American 1547-1 requirements

#Address Register address (decimal)	#Address Register address (hex)	English Name	#R/W	#Type	#Size	#SF	#Units	Range	Note
40313	0x9D79	Device RTC Time Year/Month	RW	U16	1	1	N/A	[0, 0]	MSB Year/LSB Month 13-99/1-12
40314	0x9D7A	Device RTC Time Day/Hour	RW	U16	1	1	N/A	[0, 0]	MSB Year/LSB Month 13-99/1-12
40315	0x9D7B	Device RTC Time Minute/Second	RW	U16	1	1	N/A	[0, 0]	MSB Year/LSB Month 13-99/1-12
40323	0x9D83	LVRT switch	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable
40324	0x9D84	HVRT switch	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable
40325	0x9D85	Islanding Disable	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable
40326	0x9D86	Shadow mode switch	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable

40327	0x9D87	Software ARP Switch	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable
40328	0x9D88	Arp Load Limit Set	RW	U32	2	1	W	[0,1000000]	Specific power value
40330	0x9D8A	Power On	RW	U16	1	1	N/A	[0, 0]	Write 0 to take effect
40331	0x9D8B	Power Off	RW	U16	1	1	N/A	[0, 0]	Write 0 to take effect
40332	0x9D8C	Restart	WO	U16	1	1	N/A	[0, 0]	Write 0 to take effect
40336	0x9D90	Set the ARP power percentage	RW	U16	1	1	%	[0, 500]	Percent*nominal power
40345	0x9D99	Hardware ARP Switch	RW	U16	1	1	N/A	[0,1]	For Australian safety regulations 0:disable 1:enable
40350	0x9D9E	N-PE Drtector Switch	RW	U16	1	1	N/A	[0,1]	0:disable(Default) 1:enable
40351	0x9D9F	N-PE Fault Limit	RW	U16	1	10	V	[0,2300]	Default 50V
40352	0x9DA0	Shadow mode switch	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable
40362	0x9DAA	Shadow mode switch	RW	U16	1	1	N/A	[0, 1]	0:disable 1:enable
40365	0x9DAD	GFCI Fault Clear Enable	RW	U16	1	1	N/A	[0,1]	0:No action 1:Clear GFCI fault
40480	0x9E20	Active Control adjust	RW	U16	1	1	%	[0,100]	Corresponding to 0%-100% active power
40481	0x9E21	PF reactive power adjust	RW	U16	1	1	N/A	[0, 0]	[1,20], [80,100] 1-20,lagging 0.99-0.8 80-100,leading 0.80-1, Note: if read command returns to 0xffff, it means that the inverter is not in PF setting mode

40482	0x9E22	Reactive power adjust	RW	S32	2	1	Var	[0, 0]	accurate value if read command returns to 0xffffffff, it means that the inverter not in reactive power setting mode.
40484	0x9E24	Reactive power adjust (percentage adjust)	RW	S16	1	1	%	[-60,60]	Percent*Nominal power
40500	0x9E34	Safety Country Code	RW	U16	1	1	N/A	[0, 65535]	
42400	0xA5A0	Fix Q switch	RW	U16	1	1	N/A	[0,1]	0: off 1: enable (1547 added)
42401	0xA5A1	Fix PF switch	RW	U16	1	1	N/A	[0,1]	0: off 1: enable (1547 added)
42402	0xA5A2	Fix PF Lead/Lag	RW	U16	1	1	N/A	[0,1]	0: Lead 1: Lag (1547 added)

Note:

- **U16: 16 bit unsigned number**
- **S16: 16 bit signed number**
- **U32: 32 bit unsigned number**
- **S32: 32 bit signed number**
- **RO:read only WO:write only R/W:read & write**
- **Register setting range column [0,0] indicates that no range limit is set. If there are special instructions in the remarks, the instructions shall prevail**
- **Register setting range represents the data range that communication can fill in, the actual setting range may be slightly different according to different models. The data result of the frame returned by MODBUS writing data shall prevail.**

Table 8-2: Error Code

Bit NO	Dec	Hex	Error message	Description
Bit31	2147483648	0x80000000	SCI Fail	1. Temporary phenomenon caused by external factors (such as the influence of magnetic field, etc.) 2. 2. There is a problem with the control board
Bit30	1073741824	0x40000000	Flash R/W Fail	3. Temporary phenomena caused by external factors (such as the influence of magnetic fields, etc.) 2. Internal damage to the machine
Bit29	536870912	0x20000000	Grid frequency overrun	1. Incorrect safety regulation settings 2. The grid frequency is unstable
Bit28	268435456	0x10000000	AFCI Fault	1. Poor contact of pv strings 2. Abnormal insulation between PV string and ground
Bit27	134217728	0x08000000	TBD	TBD
Bit26	67108864	0x04000000	TBD	TBD
Bit25	33554432	0x02000000	Relay Check Fail	1. Relay abnormality 2. Control circuit abnormality 3. AC side wiring abnormality (there may be a virtual connection or short circuit)
Bit24	16777216	0x01000000	TBD	TBD
Bit23	8388608	0x00800000	Export Power Limit Fault -Hardware	Abnormal Export Power Limit function (Australian safety regulations)
Bit22	4194304	0x00400000	TBD	TBD
Bit21	2097152	0x00200000	TBD	TBD.
Bit20	1048576	0x00100000	TBD	TBD
Bit19	524288	0x00080000	High DC component	The machine detects that the internal DC input component is outside the normal range.
Bit18	262144	0x00040000	Isolation Fail	1. The ground wire of the pv panel is not connected or is connected incorrectly 2. The DC wire is damaged 3. The AC terminal Neutral and ground wire are incorrectly connected 4. The error occurs in the morning and evening or in rainy day, and the air humidity is relatively high, which can easily cause ISO Report an error
Bit17	131072	0x00020000	Vac Fail	Grid voltage out of tolerable range
Bit16	65536	0x00010000	FAN Fail	The external fan failure
Bit15	32768	0x00008000	PV Over Voltage	PV Voltage is too high
Bit14	16384	0x00004000	TBD	TBD
Bit13	8192	0x00002000	Over temperature	Temperature is too high
Bit12	4096	0x00001000	TBD	TBD
Bit11	2048	0x00000800	DC Bus High	DC bus is too high

Bit10	1024	0x00000400	Ground I Fail	Ground current is too high
Bit9	512	0x00000200	Utility Loss	Utility is unavailable
Bit8	256	0x00000100	TBD	TBD
Bit7	128	0x00000080	TBD	TBD.
Bit6	64	0x00000040	TBD	TBD
Bit5	32	0x00000020	TBD	TBD
Bit4	16	0x00000010	TBD	TBD
Bit3	8	0x00000008	TBD	TBD
Bit2	4	0x00000004	TBD	TBD.
Bit1	2	0x00000002	AC HCT Fail	The output current sensor is abnormal
Bit0	1	0x00000001	GFCI Fail	GFCI check failure

9. Example

9.1 Set the active power adjustment value (percentage)

Host computer sends

Device address	write function	Register starting address		Number of registers	Number of bytes	data		CRC check	
F7H	10H	9EH	20H	00H, 01H	02H	00H	14H	F1 H	52H

It means setting the active power adjustment value to 20, which means setting the upper limit of active power output to 20% of rated power.

Slave device return

Device address	write function	Register starting address		Number of registers	CRC check	
F7H	10H	9EH	20H	00H, 01H	3BH	7DH

9.2 Read the active power adjustment value (percentage)

Host computer sends

Device address	Read function	Register starting address		Number of registers	CRC check	
F7H	03H	9EH	20H	00H, 01H	BH	BH

Slave device return

Device address	Read function	Number of bytes	Data high bit	Data low bit	CRC check	
F7H	03H	02H	00H	14H	70H	5EH

It means that the active power adjustment value is set to 20, which means the upper limit of the

active power is set to 20% of rated power.

9.3 Telemetry active power

Host computer sends

Device address	Read function	Register starting address		Number of registers	CRC check	
F7H	03H	75H	AFH	00H,02H	FAH	H

Slave device return

Device address	Read function	Number of bytes	High 16 bits		Low 16 bits		CRC check	
F7H	03H	04H	00H	00H	04H	4C H	6FH	09H

Indicates that the current output active power is 1110W.

9.4 Set the reactive power adjustment value (reactive power value)

Host computer sends

Device address	write function	Register starting address		Number of registers	Number of bytes	High 16 bits		Low 16 bits		CRC check	
F7H	10H	9EH	22H	00H,02H	04H	00H	00H	0CH	94H	89 H	2C H

Set the reactive power adjustment value to 3220Var .

Slave device return

Device address	write function	Register starting address		Number of registers	CRC check	
F7H	10H	9EH	22H	00H, 02H	DH	BH

9.5 Read the reactive power adjustment value (reactive power value)

Host computer sends

Device address	Read function	Register starting address		Number of registers	CRC check	
F7H	03H	9 EH	22H	00H,02H	5FH	7FH

Slave device return

Device address	Read function	Number of bytes	High 16 bits		Low 16 bits		CRC check	
F7H	03H	04H	00H	00H	0CH	94H	68H	93H

Indicates that the reactive power adjustment value is set to 3220Var.

9.6 Set the reactive power adjustment value (percentage)

Host computer sends

Device address	write function	Register starting address		Number of registers	Number of bytes	data bits		CRC check	
F7H	10H	9EH	24H	00H, 01H	02H	00H	32H	71H	0CH

Set the reactive power adjustment value to 50, which means the reactive power output is 50 % of rated power

Slave device return

Device address	write function	Register starting address		Number of registers	CRC check	
F7H	10H	9EH	24H	00H, 01H	7AH	BH

9.7 Telemetry reactive power

Host computer sends

Device address	Read function	Register starting address		Number of registers	CRC check	
F7H	03H	75H	B7H	00H,02H	7AH	B7H

Slave device return

Device address	Read function	Number of bytes	High 16 bits		Low 16 bits		CRC check	
F7H	03H	04H	00H	00H	07H	d8H	6EH	56H

Indicates that the current output reactive power is 2008Var

9.8 Set PF adjustment value

Host computer sends

Device address	write function	Register starting address		Number of registers	Number of bytes	data		CRC check	
F7H	10H	9EH	21H	00H,01H	02H	00H	5AH	70H	B7H

Used to set the power factor adjustment value to 90, which is +0.9 .

Slave device return

Device address	write function	Register starting address		Number of registers	CRC check	
F7H	10H	9EH	21H	00H,01H	6AH	BDH

9.9 Read the PF adjustment value

Host computer sends

Device address	Read function	Register starting address		Number of registers	CRC check	
F7H	03H	9EH	21H	00H,01H	EFH	7EH

Slave device return

Device address	Read function	Number of bytes	Data high bit	Data low bit	CRC check	
F7H	03H	02H	00H	5AH	H	6A H

The obtained PF adjustment value is 90 which means the PF is +0.9.